

PUBH7440 Homework 2

alój7980

February 2025

1 Problem 1

The conjugate prior for the binomial distribution is the Beta distribution.

$$P(X|\theta) = \binom{n}{X} \theta^X (1-\theta)^{n-X}$$

$$P(\theta) = \frac{\theta^{\alpha-1} (1-\theta)^{\beta-1}}{B(\alpha, \beta)}$$

$$P(\theta|X) \propto P(X|\theta)P(\theta)$$

$$\propto \theta^X (1-\theta)^{n-X} \cdot \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

$$\propto \theta^{X+\alpha-1} (1-\theta)^{n-X+\beta-1}$$

$$P(\theta|X) \sim \text{Beta}(\alpha + X, \beta + n - X)$$

In this Beta distribution, α represents the count of successes and β represents the count of failures for which the prior is given weight when compared to the actual successes and failures in the data. A relatively non-informative prior would have low values of both parameters.

2 Problem 2

The conjugate prior for the negative binomial distribution is the Beta distribution.

$$P(X = k|p) = \binom{k+r-1}{r-1} p^r (1-p)^k, \quad k \geq 0.$$

$$P(p) = \frac{p^{\alpha-1} (1-p)^{\beta-1}}{B(\alpha, \beta)}$$

$$\begin{aligned}
P(p|X) &\propto P(X|p)P(p) \\
&\propto p^r(1-p)^k \cdot p^{\alpha-1}(1-p)^{\beta-1} \\
&\propto p^{r+\alpha-1}(1-p)^{k+\beta-1}
\end{aligned}$$

$$P(p|X) \sim \text{Beta}(\alpha + r, \beta + k)$$

In this Beta distribution, α represents the count of successes and β represents the count of failures for which the prior is given weight when compared to the actual successes and failures in the data. A relatively non-informative prior would have low values of both parameters.

3 Problem 3

The conjugate prior for the multinomial distribution is the Dirichlet distribution.

$$\begin{aligned}
P(\boldsymbol{\theta}|\mathbf{X}) &\propto P(\mathbf{X}|\boldsymbol{\theta})P(\boldsymbol{\theta}). \\
P(\boldsymbol{\theta}|\mathbf{X}) &\propto \left(\prod_{i=1}^K \theta_i^{X_i} \right) \cdot \left(\prod_{i=1}^K \theta_i^{\alpha_i-1} \right). \\
P(\boldsymbol{\theta}|\mathbf{X}) &\propto \prod_{i=1}^K \theta_i^{X_i+\alpha_i-1}.
\end{aligned}$$

$$P(\boldsymbol{\theta}|\mathbf{X}) \sim \text{Dirichlet}(\alpha_1 + X_1, \dots, \alpha_K + X_K).$$

In this Dirichlet distribution, the α_i parameters represent the counts of observing each outcome i whose probability is represented by θ_i . The way to make this a relatively non informative prior is by using small values for all of the α_i s.

4 Problem 4

The conjugate prior for the Gamma distribution with known α is the Inverse Gamma distribution.

$$\begin{aligned}
P(\beta|X) &\propto P(X|\beta)P(\beta). \\
P(\beta|X) &\propto (\beta^\alpha e^{-\beta X}) \cdot \left(\beta^{-\nu-1} e^{-\frac{\lambda}{\beta}} \right). \\
P(\beta|X) &\propto \beta^{\alpha-\nu-1} e^{-\beta X} e^{-\frac{\lambda}{\beta}}.
\end{aligned}$$

$$P(\beta|X) \sim \text{Inverse-Gamma}(\nu + \alpha, \lambda + X).$$

For this inverse gamma distribution, α controls the heaviness of the tail and β controls the spread of the distribution. The way to make this a relatively non informative prior is by using very small values for both parameters.

5 Problem 5

The conjugate prior for the Gamma distribution with known β is the Gamma distribution with known β .

$$P(\alpha) = \frac{\lambda^\nu}{\Gamma(\nu)} \alpha^{\nu-1} e^{-\lambda\alpha}.$$

$$P(\alpha|X) \propto P(X|\alpha)P(\alpha).$$

$$P(\alpha|X) \propto (\beta^\alpha X^{\alpha-1}) \cdot (\alpha^{\nu-1} e^{-\lambda\alpha}).$$

$$P(\alpha|X) \propto \alpha^{\nu-1} e^{-\lambda\alpha} \cdot \beta^\alpha X^{\alpha-1}.$$

$$P(\alpha|X) \sim \text{Gamma}(\nu + 1, \lambda - \log \beta + \log X).$$

For this gamma distribution, α controls the heaviness of the tail and β controls the rate of decay. The way to make this a relatively non informative prior is by using very small values for both parameters.